Prime Divisibility of Descending Concatenations in Arbitrary Bases and OEIS A077183–A077185

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Abstract

Let $\mathcal{R}_b(k)$ denote the base-b integer obtained by concatenating $k, k-1, \ldots, 2, 1$ in descending order. We prove that a prime p divides at least one such integer if and only if $p \nmid b$, and that for every prime $p \nmid b$ infinitely many endpoints k exist. For $p \nmid b(b-1)$ the proof gives an explicit construction with $k < b^{2r+1}$, where $r = \mathrm{ord}_p(b)$, using a nonzero affine translation over $\mathbb{F}_p$. In decimal this proves the 2005 positivity conjecture for OEIS A077183 and the associated A077185 quotient sequence: the only zero terms occur at prime indices 1 and 3. All assertions are independent of finite searches.

1 Definitions and main theorem

For an integer base $b \geq 2$, write $\ell_b(t)$ for the number of base-b digits of $t \geq 1$, and define

$$\mathcal{L}_b(k) = \sum_{t=1}^{k} \ell_b(t), \qquad \mathcal{R}_b(k) = (k)_b \| (k-1)_b \| \cdots \| (1)_b,$$

where the concatenated digits are interpreted as a single integer. Set $\mathcal{R}_b(0) = 0$ and $\mathcal{L}_b(0) = 0$. Thus

$$\mathcal{R}_b(k) = \mathcal{R}_b(k-1) + k\, b^{\mathcal{L}_b(k-1)} \quad (k \geq 1). \tag{1}$$

OEIS A077183 asks for the least k such that the n-th prime divides $\mathcal{R}_{10}(k)$, and records zero when there is no such k. Its companion A077185 is the quotient of this smallest concatenation by the same prime [1, 2, 3]. The 2005 conjecture asserts that A077183 has zero terms only at the indices of 2 and 5.

Theorem 1 (Prime dichotomy in every base)**.** *Let $b \geq 2$ be an integer and p be a prime. Then*

$$\big(\exists k \geq 1 :\ p \mid \mathcal{R}_b(k)\big) \quad \Longleftrightarrow \quad p \nmid b.$$

When $p \nmid b$, infinitely many distinct such k exist. If $p \nmid b(b-1)$ and $r = \mathrm{ord}_p(b)$, a solution may be chosen satisfying

$$1 \leq k \leq b^{2r} - 1 + (p-1)r < b^{2r+1}. \tag{2}$$

2 A constant, nonzero translation

Lemma 1 (Fixed-width period step)**.** *Suppose $p \nmid b(b-1)$ and put $r = \mathrm{ord}_p(b)$. Let $d \equiv 1 \pmod r$, $L = b^{d-1}$, and $A = b^{\mathcal{L}_b(L-1)} \pmod p$. Whenever the integers $L, L+1, \ldots, L+jr-1$ all have exactly d digits,*

$$\mathcal{R}_b(L-1+jr) \equiv \mathcal{R}_b(L-1) + jAr(b-1)^{-1} \pmod p. \tag{3}$$

The coefficient of j is nonzero modulo p.

Proof. Write $k = L + i$, where $0 \leq i$ and k has d digits. The exponent in (1) is

$$\mathcal{L}_b(k-1) = \mathcal{L}_b(L-1) + di.$$

Because $d \equiv 1 \pmod r$ and $b^r \equiv 1 \pmod p$, we have $b^{di} \equiv b^i \pmod p$. Consequently, after adding jr new blocks,

$$\mathcal{R}_b(L-1+jr) - \mathcal{R}_b(L-1) \equiv A \sum_{i=0}^{jr-1} (L+i) b^i \pmod p. \tag{4}$$

The standard geometric identities give

$$\sum_{i=0}^{r-1} b^i \equiv 0, \qquad \sum_{i=0}^{r-1} i b^i \equiv \frac{r}{b-1} \pmod p. \tag{5}$$

For completeness, $b^r \equiv 1$ implies $\sum_{i=0}^{r-1} b^i = (b^r - 1)/(b-1) \equiv 0$; applying a finite geometric derivative gives

$$\sum_{i=0}^{r-1} i b^i = \frac{b - rb^r + (r-1)b^{r+1}}{(1-b)^2} \equiv \frac{r}{b-1} \pmod p.$$

In each consecutive r-term group of (4), $b^{i+r} \equiv b^i$. The starting offset increases by r, but its contribution vanishes because $\sum b^i \equiv 0$. Thus every group contributes exactly $Ar/(b-1)$, proving (3). Since $p \nmid b$, $A \neq 0$; also $r \mid p-1$ and hence $p \nmid r$. Finally $p \nmid b-1$, so $Ar/(b-1) \neq 0 \pmod p$. □

3 Proof and explicit witness

Proof of Theorem 1. First, if $p \mid b$, then (1) yields $\mathcal{R}_b(k) \equiv \mathcal{R}_b(k-1) \equiv \cdots \equiv 1 \pmod p$; there is no solution. If $p \mid b-1$, then $b \equiv 1 \pmod p$, so every concatenation obeys

$$\mathcal{R}_b(k) \equiv 1 + 2 + \cdots + k = \frac{k(k+1)}{2} \pmod p.$$

In particular, for every $s \geq 1$, the choice $k = 2ps$ is a solution, proving infinitude in this case (also when $p = 2$).

It remains to suppose $p \nmid b(b-1)$. Write $r = \operatorname{ord}_p(b)$, take $d = 2r+1$, and set

$$L = b^{2r}, \quad m_0 = L - 1, \quad A = b^{\mathcal{L}_b(m_0)} \pmod p, \quad \Delta = A\, r(b-1)^{-1} \pmod p.$$

Here $r \mid p-1$, so Δ is invertible modulo p. Choose j as the least nonnegative residue satisfying

$$\boxed{j \equiv -\mathcal{R}_b(m_0)\Delta^{-1} \pmod p, \quad 0 \leq j < p.} \tag{6}$$

Because $p \mid b^r - 1$, we have $p < b^r$; also $r < (b-1)b^r$ for all $b \geq 2, r \geq 1$. Therefore

$$(p-1)r < (b-1)b^{2r},$$

so the integers $L, \ldots, L - 1 + (p-1)r$ remain inside the d-digit range $[L, bL-1]$. Applying Lemma 1,

$$\mathcal{R}_b(m_0 + jr) \equiv \mathcal{R}_b(m_0) + j\Delta \equiv 0 \pmod p.$$

The displayed bound (2) follows from $j \leq p-1$.

For infinitely many further solutions, repeat the construction with $d = 1 + tr$ for any sufficiently large integer t. The same constant translation applies, and the d-digit window eventually contains all p candidates. The resulting endpoints have unbounded digit lengths, hence are distinct. □

Corollary 1 (Resolution of A077183 and A077185)**.** *For all $n \geq 1$,*

$$\boxed{\text{A077183}(n) > 0 \iff n \notin \{1, 3\}.}$$

The displayed zero values of A077185 are precisely at indices 1 and 3; all other terms are well-defined positive integers. For every prime $p \geq 7$ the least endpoint in A077183 is bounded by (2) with $b = 10$ and $r = \operatorname{ord}_p(10)$.

Proof. Apply Theorem 1 with $b = 10$. The prime divisors of the base are exactly 2 and 5. The exception $p = 3$ is covered by $3 \mid 10 - 1$; indeed $\mathcal{R}_{10}(2) = 21$. The index of prime 2 is 1 and that of prime 5 is 3. Since A077185 is the corresponding quotient by a positive prime, its terms are positive exactly at the nonexceptional indices. □

4 Verification and perspective

The accompanying Python script independently reproduces the first 63 published A077183 endpoints and the first 12 A077185 quotient values; it constructs and verifies the modular witness for primes up to 101 (and selected additional bases). It never builds the astronomically large concatenations for these constructive witnesses: modular geometric sums evaluate the blocks exactly. The symbolic proof, not the finite calculations, establishes the result.

The proof concerns existence, not a closed form for the *least* endpoint or the associated quotient. Its mechanism differs from the ascending or cyclic-digit-rotation Champernowne constructions: the natural recurrence here prepends an integer block, and the nonzero period increment comes directly from a weighted geometric sum.

References

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